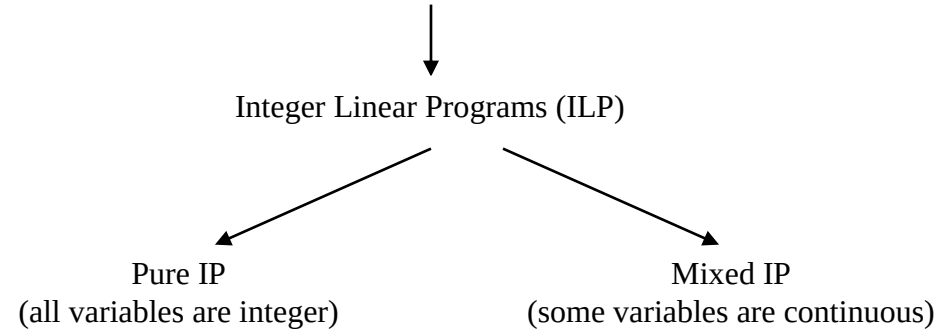
INTEGER LINEAR PROGRAMMING

LP's in which some or all variables are restricted to take integer values



Modeling Examples

1) Capital Budgeting

Decisions about whether or not to undertake a project is usually made under limited-budget considerations and preset priorities.

Ex:

Ex:

Add the following constraint to the above problem:

Ex:

Five items are to be loaded in a vessel. Weight, volume and value for each item are given in table below. Available cargo weight and volume are 112 tons and 109 m^3, respectively. Formulate the ILP model.

2) Fixed Charge Problem

The initial tooling of a machine prior to starting production incurs a fixed setup cost regardless of how many units are manufactured. Once the setup is done, the cost of labor and material is proportional to the amount produced.

That is:

Ex: Jobco is planning to produce at least 2000 widget on three machines. The pertinent data of the situation is as follows:

Machine	Setup cost	Pr. Cost/unit	Capacity
1	300	2	600
2	100	10	800
3	200	5	1200

Bild the ILP model.

Solution Algorithms for ILP

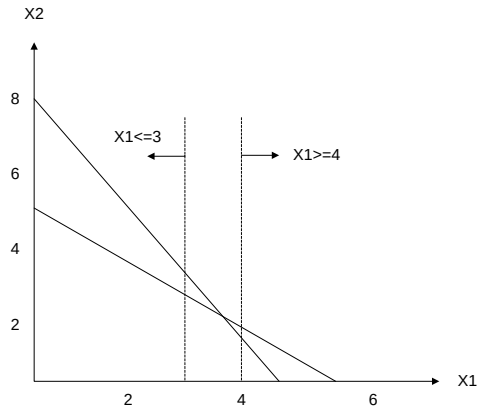
Two main methods:

1) Branch-and-Bound (B&B) Algorithm
2) Cutting Plane Method

B&B Algorithm

$$\begin{aligned} \max \quad & z = 5x_1 + 4x_2 \\ s.t. \quad & x_1 + x_2 \le 5 \\ & 10x_1 + 6x_2 \le 45 \end{aligned}$$

x_1, x_2 non-negative integer



First solve the problem as an LP (i.e. $x_1, x_2 \geq 0$)

LPO;

→ $x_1 = 3.75$, $x_2 = 1.25$, $z = 23.75$ (is an upper bound for the ILP)

Then choose one of the non-integer variables

Say, we select x_1 (x_1 is the branching variable)

$x_1 \leq 3$ or $x_1 \geq 4$

Two separate LP's arise;

Two questions:

1) Could we have selected x_2 as the first branching variable → Yes
2) When solving the next subproblem, could we have selected LP2 first → Yes

Summary of the B&B Algorithm (max.)

LB = -∞, i = 0

Step 1 (Fathoming / bounding)

Select LP1 the next subproblem to be examined.

Solve it

Fathom it if one of the three conditions is met:

a) Its z-value cannot yield a better value than the current LB.

b) It yields an all integer solution better than the current LB → update LB

c) It has no feasible solution

If LPi is not fathomed go to step 2

If all subproblems are fathomed, stop, the optimum is the current LB.

Step 2 (Branching)

Select one of the non-integer variable which is required to be an integer.

x_j^* current optimum value of x_j

Create two problems having the additional constraints of;

$x_j \le \lfloor x_j^* \rfloor$ and

$x_j \ge \lfloor x_j^* \rfloor + 1$

set $i = i + 1$, go to step 1.

Alternative strategies for choosing the next

Branching variable	Subproblem to be solved
- variable having the lowest index	- breadth first
- variable closest to an integer	- depth first

No strategy is shown to be strictly superior to the others.

Ex:

$$\left.\begin{array}{ll} \max & z = 18x_1 + 14x_2 + 8x_3 \\ s.t. & 15x_1 + 12x_2 + 7x_3 \le 43 \end{array}\right\}$$ 4 iter to find the optimum

29 iter to prove optimality

Cutting Plane Algorithm

$$\begin{array}{ll} \max & z = 7x_1 + 10x_2 \\ s.t. & -x_1 + 3x_2 \le 6 \\ & 7x_1 + x_2 \le 35 \\ & x_1, x_2 \ge 0 \text{ and int}. \end{array}$$

